

On the controls of preferential flow in soils of different hillslope position and lithological origin

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Abstract

Soils derived from different lithologies and their controls on preferential flow remain underexplored in forested landscapes. In the same lithology, the propensity for preferential flow occurrence at different hillslope positions also remains largely elusive. By utilizing a soil moisture response time method, we compared preferential flow occurrence between a shale site (Shale Hills, silt loam soils) and a sandstone site (Garner Run, sandy loam soils) at four hillslope positions: ridge-top, North- and South-facing mid-slopes and toe slope, for over 2 years. The catchments are neighbouring and covered by temperate forest. For the four hillslope positions, Shale Hills had higher preferential flow frequencies compared to Garner Run. Between these two catchments, the South-facing mid-slope sites showed the highest contrasts in preferential flow frequency (33.5% of events at Shale Hills vs. 8.8% at Garner Run) while the ridge-top sites showed the lowest contrasts (18.7 vs. 13.2%). Additionally, over the unfrozen period, for seven out of eight monitoring sites, drier antecedent conditions tended to be more favourable for preferential flows to occur, with significant ($p < .01$) relationships at two sites. Except for the South-facing mid-slope sites, both Shale Hills and Garner Run had two preferential flow pathways. The characteristic preferential flow pathways at Shale Hills were the B_w and C horizons, and for Garner Run, preferential flow moved from the E/AE horizon to the B_w horizon. This study shows that shale-derived soils tended to have higher preferential flow occurrence than sandstone soils, but hillslope positions exhibit different levels of contrasts. More effort should be paid to study the impact of lithology on preferential flows in the context of land surface modelling and biogeochemical reactions to improve ecosystem services of headwater catchments.

KEYWORDS

hillslope hydrology, lithology, macropore, preferential flow, sensor network, soil moisture, soil structure, subsurface lateral flow

1 | INTRODUCTION

Preferential flows are nonequilibrium channelling flows which move faster than uniform flow in the subsurface (Beven & Germann, 1982). Preferential flows occur from core to catchment scales with

corresponding mathematical formulations (Hendrickx & Flury, 2001). Among different types of preferential flows (Lin, 2010), macropore flows and subsurface lateral flows are commonly seen in forested landscapes. Macropores could be formed by tree roots (Luo et al., 2019), fauna activities (Cheng, Ogden, & Zhu, 2017) and soil

cracks (Bachmair, Weiler, & Nützmann, 2009). Macropore flows are often associated with higher water flux in a hydraulic disequilibrium state (Nimmo, 2012). Subsurface lateral flows, on the other hand, are often driven by gravitational gradients. They often flow along impermeable features such as soil-bedrock interfaces (Appels, Graham, Freer, & McDonnell, 2015; Sohr, Ries, Sauter, & Lange, 2014) and dense soil fragipans. Soil pipes are another feature that can channel subsurface flow (Holden, 2004). These pipes can be as long as tens of metres in hillslopes in temperate regions (Chappell, 2010) and can sometimes contribute up to 50% of storm flow discharge (Jones, 2010). Both macropore flow and subsurface lateral flow significantly impact catchment-scale hydrological processes and the partitioning of local water balance. However, upscaling preferential flow occurrence in complex landscapes with differing lithologies is a scientific challenge (Band et al., 2014; Liu et al., 2020).

Lithology plays an important role in controlling vadose zone water movement (Brantley, Goldhaber, & Ragnarsdottir, 2007). Soils form in different lithologies via contrasting physical and geochemical weathering processes. Conversion from primary minerals (e.g., feldspar and micas) to secondary minerals (e.g., silicate clays) defines soil texture. At later stages of pedogenesis, the arrangement and aggregation of primary soil particles into secondary units or peds defines soil structure. Soil structure and soil texture are a paired control on porosity and pore connectivity. Preferential flow usually flows through large soil pores (macropores) that have sufficient porous connectivity. These fast-flow pathways can sometimes extend all the way down to the underlying saprock (Anderson, Dietrich, Torres, Montgomery, & Loague, 1997) due to seepage pressure and permeability contrasts (Hencher, 2010). Currently, there are two knowledge gaps existing in understanding the impact of soils derived from different lithologies on preferential flows. Studies on the impact of soil textures on preferential flows are often lab-based and the special soil structures (e.g., rock fragments and orientations) cannot be precisely reconstructed in lab. Using the Brilliant Blue dye staining method, Grant, Macrae, and Ali (2019) found that preferential flows moved deeper in clay soils compared to silt loam soils. More field studies are needed to compare the relative frequency of preferential flows in soils with different textures. Furthermore, boulder fields in the mid-Appalachian Mountains have been relatively unexamined in terms of soil water relationships across contrasting lithologies. Traditional infiltration equipment, including double-ring infiltrometers and dual-head infiltrometers, requires sufficient soil depth for ring insertion in order for water to infiltrate vertically. In boulder fields, however, this is often hindered by underlying boulders that tend to diffuse water horizontally.

Hillslope position has been widely investigated in terms of its relationship with water table depth (Fan, Li, & Miguez-Macho, 2013). However, its impact on preferential flows is less clear. Using a rainfall simulator and dye staining method in a semiarid catchment, Öhrström et al. (2002) found that preferential flows occurred more at nose and hollow regions than on planar hillslopes, and the variability of preferential flows increased with increased spatial scale. In humid-temperate regions, Holden (2009) also found a strong impact of hillslope position on preferential flows. This impact, based on their conclusion, was

mainly driven by the different relationships between macropore flow and soil saturated hydraulic conductivity (K_s). Despite its importance, limited studies have tested hillslope position impact on preferential flows in shale and sandstone lithologies.

Soil moisture is a key variable that links water, energy, and biogeochemical cycles (Seneviratne et al., 2010), and controls infiltration processes (Blume, Zehe, & Bronstert, 2009; Germann & Beven, 1981). In recent years, by inferring the lag time of soil moisture responses at different depths, preferential flows were calculated and compared between different landscape positions, topographies and land uses (Demand, Blume, & Weiler, 2019; Graham & Lin, 2011; Hopkins, Gall, & Lin, 2016; Lin & Zhou, 2008; Liu & Lin, 2015; Wiekenkamp, Huisman, Bogen, Lin, & Vereecken, 2016). Traditional methods studying preferential flows, like the dye staining method (Wang, Chen, Lian, Fu, & Nie, 2018; Weiler & Flüher, 2004), are destructive and non-repetitive in the field. Breakthrough curve analysis (Luo, Lin, & Schmidt, 2010), through plotting solute concentration versus collection time for artificial soil columns, is also a snapshot-type analysis. Rainfall simulator methods (Cheng, Ogden, Zhu, & Bretfeld, 2018; Litt et al., 2019) are useful to measure bulk infiltration (f_b , the combination of matrix flow infiltration and preferential flow infiltration) at plot scales, but relatively hard to be implemented in high, steep and rocky regions. Compared to these methods, the soil moisture monitoring method has several advantages, including minimum disturbance to the original soil physical and chemical properties, allowance for repeatable observations during long-term monitoring, and the capacity to capture temporal dynamics of preferential flow processes on an event basis (Guo & Lin, 2018). The propensity for water to flow preferentially is often viewed as an inherent property for soils/hillslopes. Using the soil moisture dynamics method mentioned above, a preferential flow frequency index was calculated in several previous studies (Guo, Fan, Zhang, & Lin, 2018; Hopkins et al., 2016; Liu & Lin, 2015; Wiekenkamp et al., 2016). This index is determined by counting the number of preferential flow events and dividing this by the total number of precipitation events over a long monitoring period (typically 2 or 3 years). The soil moisture dynamics method assumes that preferential flow is measured by matric responses, which is acceptable here given the relative lack of soil pipes observed in these watersheds.

Few studies have investigated how soil moisture response time to precipitation events change in response to climate variability. In temperate regions, soil wetness changes significantly between growing and dormant seasons and could strongly impact soil water movement and biogeochemical reactions (Duncan, Band, Groffman, & Bernhardt, 2015). Matric potentials that drive soil water movement can sometimes differ by four orders of magnitude between dry and wet states for fine textured soils (Sławiński, 2011). During summer periods, precipitation events with higher intensities tend to trigger more faster flow events (Wiekenkamp et al., 2016). Consecutive freeze-thaw cycles of soils could also disrupt soil structure, especially for soils with higher clay contents (Oztas & Fayetorbay, 2003). Understanding the impact of hydroclimate variability on soil moisture response times is important for helping understand percolation variability between growing and dormant seasons.

In this study, we explored the impact of both hillslope position and soils derived from different lithological origins on preferential flows in the Susquehanna Shale Hills Critical Zone Observatory (SSHCZO) by utilizing a two-year soil moisture observation system that monitored soil moisture every 10 minutes from March 2017 to March 2019. We tested the hypothesis that *soils derived from sandstone lithology would have faster percolation rates and more preferential flows than soils derived from shale lithology*. A watershed underlain by shale (Shale Hills) and one by sandstone (Garner Run) were chosen for comparison. Four hillslope positions were monitored at each catchment: ridge-top, North-facing (N-F) mid-slope, South-facing (S-F) mid-slope, and toe slope (Figure 1). We aimed to answer three questions: (a) How did surface soil moisture (top 10 cm) and drainage differ between the two catchments across different seasons? (b) Which catchment had more preferential flow, Shale Hills (silt loam catchment) or Garner Run (sandy loam catchment)? (c) In which soil horizons were preferential flows most likely to occur?

2 | MATERIALS AND METHODS

2.1 | Field site

This work was conducted in two small forested and adjacent watersheds that are part of the SSHCZO. Shale Hills is a 7.9-ha, V-shaped

forested catchment that overlies the Rose Hill Shale Formation. Rose Hill Shale is organic-poor but Fe-rich (Meyer, Textoris, & Dennison, 1992). It has thin interbedded, fine-grained sandstone and extends along a large portion of the Appalachians (West, Kirby, Bierman, & Clarke, 2014). The Shale Hills catchment is characterized by steep (14 for S-F slope, 21 for N-F slope) and short hillslopes (50–100 m). The elevation of Shale Hills ranges from 259 to 310 m. Five soil types were mapped in Shale Hills: *Weikert* (shallow, well-drained), *Berks* (moderately deep, well-drained), *Rushtown* (deep, well-drained), *Ernest* (very deep, poorly to moderately drained) and *Blairton* (very deep, moderately well-drained). The soils were derived from Silurian-aged shale of the Clinton group (Folk, 1960), where thickness varies from 200 to 300 m (Brantley, Holleran, Jin, & Bazilevskaya, 2013). Texture of the soil is generally silt loam or silt clay loam (Lin, Kogelmann, Walker, & Bruns, 2006). Vegetation at Shale Hills is predominately deciduous trees including oaks and maples (Meinzer et al., 2013) with sparse understory (Brubaker, Johnson, & Kaye, 2018).

Garner Run, a rectangular-shaped watershed, is underlain by the Tuscarora Sandstone Formation (Hettinger, 2001). The Tuscarora Formation is 150-m-thick sandstone with thin interbedded olive-grey shale and is overlain by an upper Castanea Member (Del Vecchio et al., 2018). The elevation at Garner Run is the highest in Shaver's Creek watershed (Brantley et al., 2016). A dominant feature at Garner Run are colluvial boulder fields along the hillslopes, which is evidence

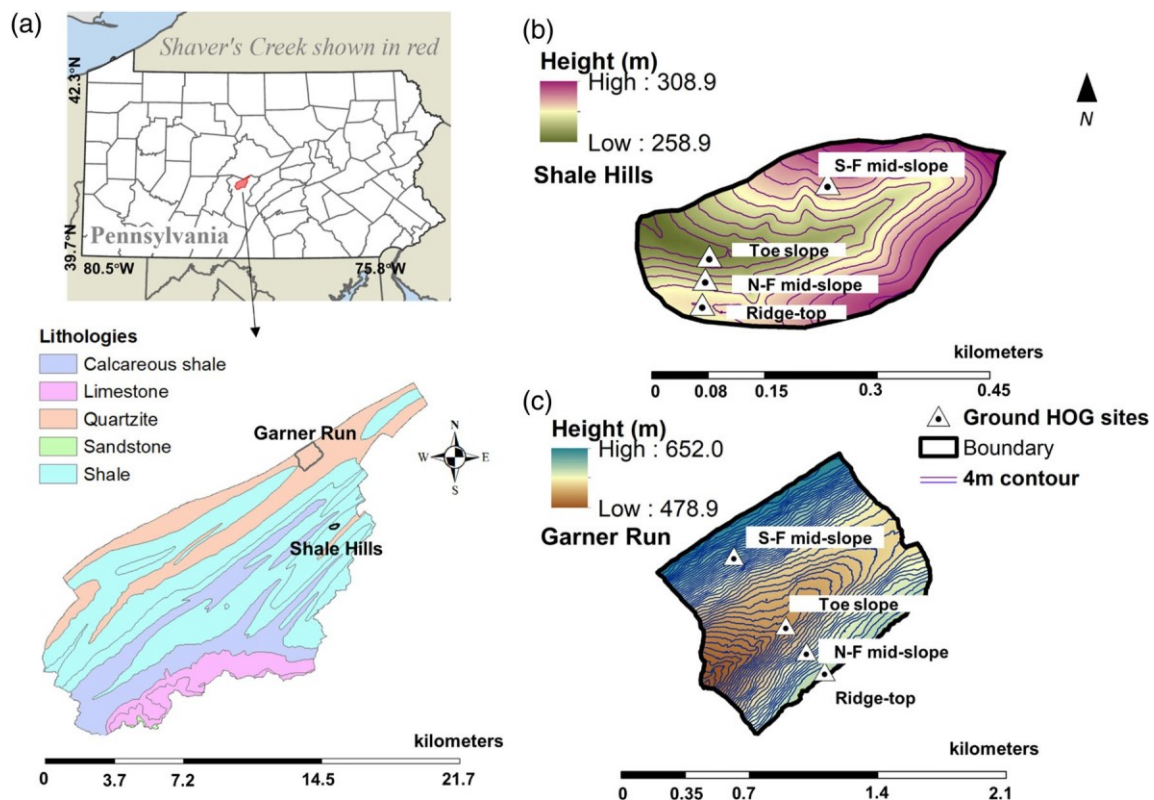


FIGURE 1 (a) Location of the study catchments; (b) Shale Hills and (c) Garner Run with soil moisture monitoring sites ("Ground HOG") plotted on

of periglacial processes (Del Vecchio et al., 2018). Like Shale Hills, Garner Run also contains a deciduous forest but with a less diverse overstory than the forest type (Brubaker et al., 2018). Compared to Shale Hills, it has less steep N-F slope (12), steeper S-F slope (17), and longer hillslopes (300–500 m). Soils at Garner Run are generally Hazleton–Dekalb series and Andover soil series, consisting of stony sandy loams, mountain slope colluvium, and poorly drained soils with high runoff regimes (Brantley et al., 2016; Hoagland et al., 2017).

The water budget has been well characterized at the Shale Hills catchment. For 2019, the maximum throughfall input was 12.2 mm 10 min^{-1} . Average throughfall input and duration per event was 12.0 mm and 16.8 hr, respectively. Total annual throughfall was 875.5 mm. Peak streamflow measured by a flume gauge located at the outlet of the watershed was $0.06 \text{ m}^3 \text{ s}^{-1}$ for dry periods and $0.25 \text{ m}^3 \text{ s}^{-1}$ for wet periods. At the Garner Run catchment, the water budget, however, was more difficult to measure (Xiao et al., 2019). This was largely due to the fact that the size of Garner Run (1.34 km^2) has a much wider boulder-strewn valley and the stream flows to one side of it. The flume therefore only captures part of the water flowing out of the catchment. During the same period, the maximum throughfall input at Garner Run was 15.8 mm 10 min^{-1} and total annual throughfall was 734.1 mm. Average throughfall input per event was 14.0 mm and average duration per event was 17.4 hr. The dry months for both catchments were from July to September due to their close locations (Figure 1).

2.2 | Hydrologic monitoring

Four hydrological observation sites were installed at both catchments (Figure 1). They were termed as “Ground HOGs” (Ground Hydrological Observation Gear). For each catchment, there was one installed at the S-F mid-slope, and the other three were installed at the N-F slope including ridge-top, mid-slope and toe slope along a catena. Related soil profile information was summarized in Table S1. The soil moisture sensors used for this study were Stevens Hydraprobe type (Stevens Water Monitoring Systems, Inc. Portland, OR), which were programmed to measure soil moisture and soil temperature continuously at a 10-min interval. Given the difficulty in extracting cores from boulder fields for gravimetric soil water content measurements, we followed recommendations from the manufacturer. The sensor calibration was kept as GENERAL, which was the default factory calibration set by Stevens and suitable for nearly all mineral soils (https://www.stevenswater.com/resources/documentation/hydraprobe/HydraProbe_Manual_Jan_2018.pdf). Thus, for this study, we used raw data except for missing or corrupted data (e.g., data less than 0 and more than 1, which was out of the range of volumetric-based soil moisture content due to power issue). At Shale Hills North-facing mid-slope site, we found jittery data for 1 month (2019 April), and we excluded it from the analysis. For each Ground HOG site, sensors were deployed at 10, 20 and 40-cm depths from the top of the mineral soil layers. A Pluvio load cell rain gauge (OTT Hydrometry) was used to measure precipitation every 10 min with a resolution of 0.01 mm. It was located on Shale Hills S-F ridge-top in an open space.

2.3 | Event-based analysis to infer infiltration rates

Precipitation buckets (TE525-WS type, Texas Electronics, resolution 0.025 mm) were deployed in September 2018 at Shale Hills N-F mid-slope, ridge-top, Garner Run N-F mid-slope, ridge-top, and S-F mid-slope, to help get precise measurements of response times. They were co-located with the soil moisture sensors and measured throughfall every 10 min. We cleaned these buckets approximately biweekly to make sure they did not get clogged by fallen leaves. Initiation of the event was defined from the time of the first tipping bucket reading when at least two pulses of rainfall were recorded by the tipping bucket in a 1-hr window. For each event, a 1% soil moisture increase was regarded as a response signal. This threshold was chosen based on past studies (Graham & Lin, 2011; Lin & Zhou, 2008; Liu & Lin, 2015) and the HydraProbe sensor properties (accuracy $\pm 3\%$, precision $\pm 0.3\%$). The threshold was set slightly higher than the noise level to avoid possible diel signal interferences (Graham & Lin, 2011). The event analysis extended from March 2019 to August 2019 and was then divided into dormant season (March–May) and growing season (June–August). The lag time between the initiation of precipitation and response signals at 10-cm and 20-cm depths were inferred as an indicator of percolation rate from soil surface to 10-cm and 20-cm depths.

2.4 | Infer preferential flows based on lag time

Preferential flow could be studied either through numerical modelling (Cheng et al., 2017; Simunek, Jarvis, van Genuchten, & Gardenas, 2003) or lab/field detection (Allaire, Roulier, & Cessna, 2009). In this study, the occurrence of preferential flow was detected by the sequence of soil moisture response time to individual precipitation events. This method was first proposed by Lin and Zhou (2008), then adopted and further developed in subsequent studies (Graham & Lin, 2011; Hardie, Lisson, Doyle, & Cotching, 2013; Hopkins et al., 2016; Liu & Lin, 2015; Wiekenkamp et al., 2016; Zhao et al., 2012). In Wiekenkamp et al. (2016), a precipitation event started when at least 1 mm precipitation was recorded within 3 hr, and ended when no precipitation was recorded within 3 hr. Given the spatial heterogeneous nature of precipitation, this method, however, might be problematic as it required precise measurement for each location. In this study, we used soil moisture change as an indicator of an event, which included both the frozen and unfrozen periods.

For each event, if water moved sequentially from the surface layer to the deeper layer, soil moisture sensors would also respond sequentially from top to the bottom. Any response scenarios differed from that would be a preferential flow event (see Figure 2 as an example of sequential and preferential flows).

Similar as the event-based analysis for percolation, we used a 1% soil moisture increase as a response signal for all three depths. Two-year data (Mar 2017–Mar 2019) was included to ensure that enough events were covered. Total event number differed from site to site due to different data gaps and minimum thresholds to trigger sensor response, but a minimum event number of 80 was ensured for each site. For each site, we calculated absolute preferential flow frequency

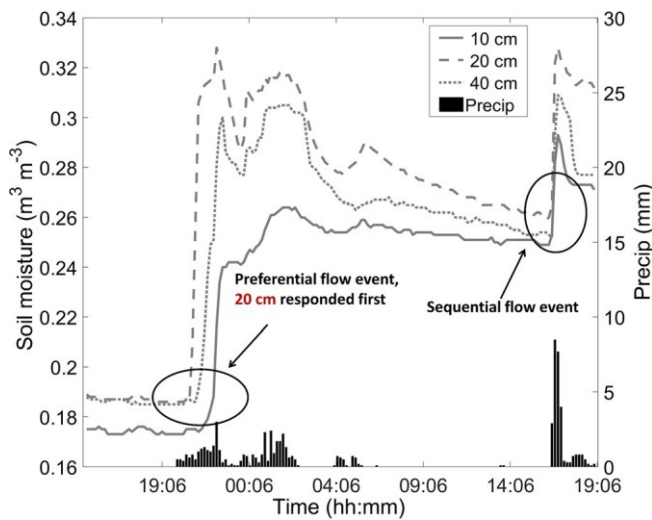


FIGURE 2 Illustration of sequential flow and preferential flow at Shale Hills South-facing mid-slope. Precipitation was measured every 10 minutes. During Event 1, under a more average precipitation with limited maximum intensity, soil moisture at 20-cm depth (dashed line) responded earlier than that at a 10-cm depth (solid line), indicating that water moved to 20 cm sooner than 10 cm. During Event 2, under a precipitation with higher maximum intensity, the three sensors responded at the same time. We categorized event 2 as a sequential flow event

and relative preferential flow frequency. Absolute preferential flow frequency was calculated by counting total preferential flow events. Relative preferential flow frequency was calculated based on the fraction of preferential flow events over the total flow events (Equation [1]).

$$PF\% = \frac{\text{Preferential flow events}}{\text{Preferential flow events} + \text{Sequential flow events}} \quad \delta 1p$$

Over the unfrozen period, we separated antecedent water content (AWC) that resulted in PF events and sequential flow events in two bins. The Shapiro-Wilk test indicated that AWC was skewed for seven out of eight sites. Log transformation of the AWC data was unable to normalize all AWC data. We therefore conducted a non-parametric Kruskal-Wallis test (Hardie et al., 2013) to compare AWC between preferential flow events and sequential flow events at a significance level of 1%. All statistical analysis was performed in Matlab (Mathwork, MA) under the Statistics and Machine Learning Toolbox.

2.5 | Preferential flow pathways at different depths

In order to link preferential flow occurrence with specific soil depths, we classified five types of preferential flow response scenarios: (a) 20 cm responded earlier than 10 cm, then followed by 40 cm; (b) 10 cm responded first, but 40 cm responded earlier than 20 cm; (c) 40 cm responded earlier than 10 and 20 cm; (d) 10 cm responded later than 20 and 40 cm; and (e) 40 cm responded earlier than 20, and 20 cm responded earlier than 10 cm. These five types of preferential

flows covered all response scenarios in our dataset. For each site, we counted the most frequent preferential flow type and the second most frequent preferential flow type. We then referred these types as first dominant preferential flow pathway and second dominant preferential flow pathway, respectively (Table 2). We also linked preferential flow occurrence depths with different horizons and in situ soil profile observations to infer linkages of soil structure with preferential flow.

3 | RESULTS

3.1 | Seasonal soil moisture pattern between Shale Hills and Garner run

During the growing season, both Shale Hills and Garner Run showed similar decreasing trends from May to July and then started to increase in August (Figure 3). For the S-F mid-slope, the median soil moisture content at the 10-cm depth decreased from 24.2 to 22.1% for Shale Hills and from 10.8 to 8.3% for Garner Run. (Figure 3). For each month, Shale Hills had a consistently higher surface soil moisture compared to Garner Run ($> 0.1 \text{ m}^3 \text{ m}^{-3}$, $p < .1$).

3.2 | Soil moisture response time difference between dormant and growing season

During the dormant season, soil moisture response time at Garner Run was similar to that at Shale Hills (Figure 4), despite the coarser soils at Garner Run (Table 1). Based on our hypothesis that Garner Run tended to have faster drainage than Shale Hills, we predicted the data points should fall beneath the 1:1 line (Figure 4), as Shale Hills would have longer response times than Garner Run for specific events. However, almost equal scattering of response times was found above and beneath the 1:1 line.

During the growing season, both Shale Hills and Garner Run had shorter response times (Figure 4) compared to the dormant season due to the higher precipitation intensities ($4.33 \text{ mm } 10 \text{ min}^{-1}$ for growing season and $1.97 \text{ mm } 10 \text{ min}^{-1}$ for dormant season). Most of the 10-cm depth response times fell above the 1:1 line, which suggested a faster percolation rate at Shale Hills. Due to sensor malfunction in 2019 August, we only had five points at the 20-cm depth, and they also fell on or above the 1:1 line.

3.3 | Lithology, hillslope position and initial wetness impact on preferential flows

Based on data analysis of soil moisture over 2 years, in terms of the relative frequency, Shale Hills showed a significantly higher preferential flow frequency for all hillslope positions compared to Garner Run (Figure 5). Comparing Shale Hills with Garner Run, preferential flow frequency differed most at S-F mid-slope (33.5 vs. 8.8%) and least at ridge-top (18.7 vs. 13.2%). The highest preferential flow frequency was found at the N-F mid-slope for both Shale Hills (53.1%) and Garner Run (47.4%). The lowest preferential flow frequency at Shale Hills

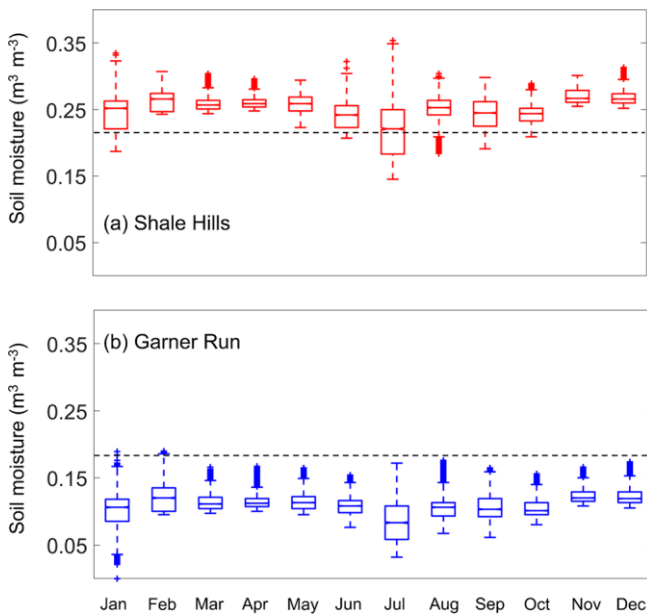


FIGURE 3 (a)–(b), Soil moisture box-and-whisker plots at the 10-cm depth from Shale Hills and Garner Run South-facing mid-slopes for year 2018. In each plot, the error bars referred to the range, the rectangles referred to the 25th and 75th percentiles, the horizontal lines referred to the median values, and the red crosses referred to the outliers. For slope position variance, readers are referred to Li et al. (2018). We plotted two dashed line on figure (a) and (b) that indicated catchment average (0.22 $\text{m}^3 \text{m}^{-3}$ for Shale Hills and 0.18 $\text{m}^3 \text{m}^{-3}$ for Garner Run) over the unfrozen period

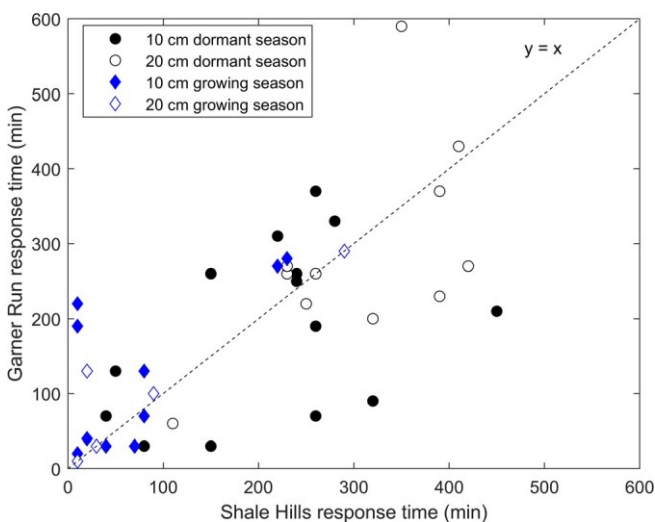


FIGURE 4 Response time between Shale Hills and Garner Run for 10 and 20-cm depths. Black dots were data from 2019 dormant season (March, April and May). Blue dots were data from 2019 growing season (June, July and August). There was no clear relationship between Shale Hills and Garner Run in terms of response time, whereas the growing season showed shorter response times compared with the dormant season due to stronger storms

was found at the ridge-top (18.7%) and for Garner Run, it was at S-F mid-slope (8.8%). For these two locations, in addition to a low

preferential flow frequency, both sites also had lower soil moisture content compared with other locations.

In terms of initial wetness, two out of four sites (N-F mid-slope and toe slope) at Shale Hills differed significantly between preferential flow events and sequential flow events (Figure 6). At Garner Run, only the toe slope site's initial soil wetness differed significantly between preferential flow events and sequential flow events. In general, drier antecedent content at Shale Hills and Garner Run was more favourable for preferential flows to occur. The only exception was the Shale Hills toe slope site where wet soil was more favourable for preferential flows to occur.

3.4 | Preferential flow pathways between Shale Hills and Garner run

At the Shale Hills ridge-top site, the dominant preferential flow pathways were the B_w horizon and part of the C horizon (Table 2; Table S1). For the same hillslope position at Garner Run, water quickly bypassed the E horizon and percolated to the B_{hs} horizon. At the N-F mid-slope site, preferential flows bypassed the B_{hs} and B_s horizons at Garner Run and B_w and C horizons at Shale Hills. On the opposite aspect, preferential flows bypassed part of the AE horizon at Garner Run and A and B_{w1} horizons at Shale Hills, respectively. At the Shale Hills toe slope site, the 40-cm depth was periodically saturated (Figure S1). A similar saturated state at the same depth was also found for Garner Run ridge-top site (Figure S1). Preferential flow moved from Ae and B_w horizons to B_s horizon for Shale Hills toe slope site, and for Garner Run, it moved mostly through the B_{w1} and B_{w2} horizons.

4 | DISCUSSION

4.1 | Spatial pattern of soil moisture between Shale Hills and Garner run

For the selected slope position (S-F mid-slope), compared to Shale Hills, Garner Run had less surface water stored, which was likely related to its coarser soil texture (larger rocks and 4.0% less clay than Shale Hills) (Table 1). Shale Hills also had higher catchment average water content than Garner Run over the unfrozen period (Figure 3).

The spatial difference of soil moisture, however, differed between Shale Hills and Garner Run, as was previously discussed in Li et al. (2018). Driven by higher gravitational gradients, the steeper N-F slope at Shale Hills likely deliver more water to the toe slope area and thus dry spots were created in the ridge area and mid-slope area. At Garner Run, Li et al. (2018) found only small differences between toe slope and ridge area along the N-F slope. However, the driest location was found on the S-F mid-slope area where higher rock fragment content was found (Table 1). The higher rock fragments would lead to less storage space for water. Thus, soil texture, slope gradient and different rock fragments co-controlled the spatial heterogeneity of soil moisture. This would then impact the infiltration rates and preferential flow occurrence at different hillslope positions.

TABLE 1 Topographic and pedologic information for different Ground HOG points at Shale Hills and Garner Run

	Elevation	Slope	Curvature	TWI ^a	Clay ^b	Rock fragments
	m	o	* 10 ⁻²		%	%
Shale Hills						
Ridge-top	284.0	1.3	-1.5	3.9	19.4	56.8
S-F mid-slope	289.4	20.6	0.2	5.3	16.5	38.1
N-F mid-slope	274.2	24.2	0.6	4.4	18.2	58.9
Toe slope	263.4	7.1	-1.2	6.0	27.4	19.5
Garner run						
Ridge-top	587.2	2.5	29.6	3.1	14.4	29.0
S-F mid-slope	596.7	15.7	12.5	2.0	12.5	67.0
N-F mid-slope	554.6	7.9	14.8	2.0	13.5	45.0
Toe slope	512.0	8.3	18.1	2.6	19.8	35.0

^aTopographic wetness index. It was calculated based on the logarithmic relationship between contributing area and local slope.

^bClay content and rock fragments for Shale Hills were adopted from Lin et al. (2006). Clay content and rock fragments were adopted from Brantley et al. (2016).

Note: Calculations were based on a 1-m digital elevation model (DEM) for both catchments.

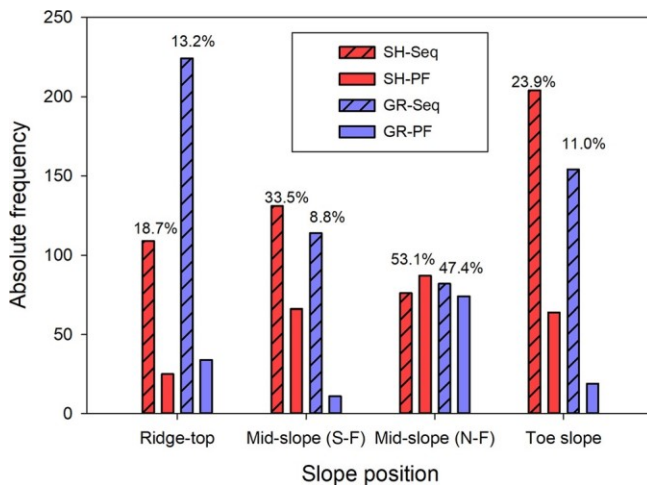


FIGURE 5 Frequency of preferential and sequential flow events for Shale Hills and Garner Run at each slope position. The relative frequency (%) of preferential flow was also noted for each slope position. SH-Seq, Shale Hills sequential flow, SH-PF, Shale Hills preferential flow, GR-Seq, Garner Run sequential flow, GR-PF, Garner Run preferential flow. For all slope positions, Shale Hills had a higher relative preferential flow frequency compared to Garner Run with the South-facing mid-slope position showing the highest contrasts

4.2 | Did preferential flows accelerate water percolation in soils?

Theoretically, soils with coarser texture had faster percolation rates compared to those with finer texture due to larger pore sizes. For porous media, Kozeny-Carman equation predicted rapid water pressure decrease rate associated with smaller pores. Thus, faster percolation rates were normally found in sandy soils where they could maintain their pressure gradients for a longer time. Field

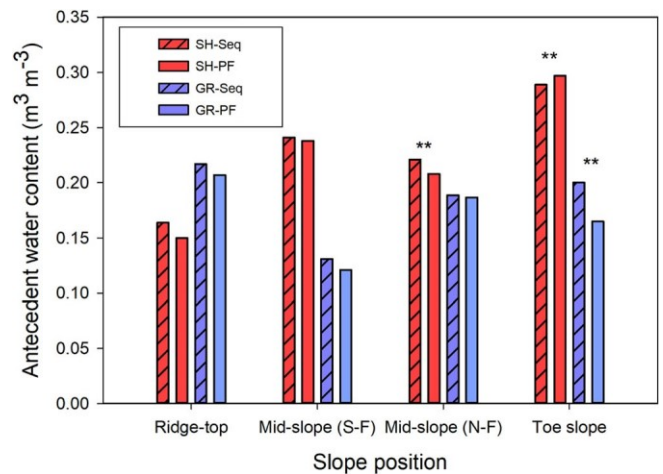


FIGURE 6 Average antecedent water content prior to preferential and sequential flow events in Shale Hills and Garner Run (**: $p < .01$ by non-parametric Kruskal-Wallis test) over the unfrozen period. Preferential flows generally were associated with drier antecedent soil moisture except for the toe slope site at Shale Hills

measured infiltration or percolation rates, however, did not always align with this relationship. In Ahmed, Gulliver, and Nieber (2015), macropores created by plant roots accelerated infiltration rate for different soil texture classes. Using a hood infiltrometer for different types of plant species, Fischer et al. (2014) did not find any correlation between soil texture and infiltration rates. The sandstone watershed, Garner Run, did not show a significantly shorter response time compared to Shale Hills (Figure 4). This suggested that water not only moved through micropores, but also moved through macropores among soil aggregates at the Shale Hills catchment. This was not predictable if we only took soil texture into consideration.

TABLE 2 Absolute frequency of preferential flow events at different landscape positions and depths at the Shale Hills (shale) and Garner Run (sandstone) catchments from 2017–2018

		20 cm earlier than 10 cm	40 cm earlier than 20 cm	40 cm earlier than 10 cm	Both 20 cm and 40 cm earlier than 10 cm	40 cm earlier than 20 cm and 20 cm earlier than 10 cm
Shale Hills	Ridge-top	11	12	1	1	0
	S-F mid-slope	35	7	3	17	4
	N-F mid-slope	17	49	11	5	5
	Toe slope	34	9	20	1	0
Garner run	Ridge-top	17	8	9	0	0
	S-F mid-slope	0	11	0	0	0
	N-F mid-slope	2	46	13	6	7
	Toe slope	3	11	4	1	0

Note: Total rainfall events varied due to different data gaps.

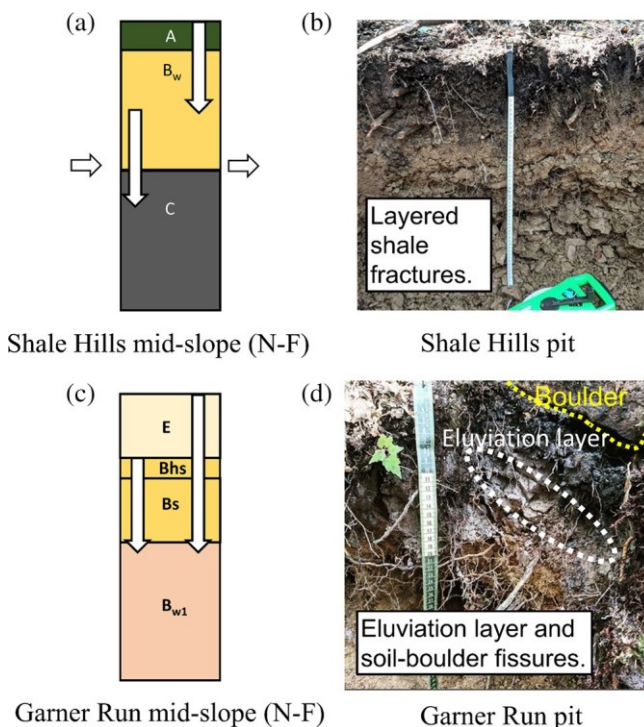


FIGURE 7 Preferential pathways for Shale Hills and Garner Run. Shale Hills horizon information was adopted from Lin et al. (2006) and Garner Run horizon information was adopted from Brantley et al. (2016). For (a) and (c), we showed potential preferential flow pathways for Shale Hills and Garner Run North-facing mid-slope position. Field observations on soil profiles between Shale Hills and Garner Run were illustrated in (b) and (d)

The higher preferential flow rate at Shale Hills might come from two possible reasons: (a) Water movement was channelled by the organized orientation of the rock fragments on top of the saprock based on field observation (Figure 7); (b) More subsurface flows occurred along the steeper slope gradient (Ribolzi et al., 2011) on top of the less weathering resistant lithology at Shale Hills. The rock

pattern on the top soil layer may likely be the best explanation for more preferential flows at Shale Hills, as past field studies at Shale Hills and nearby shale catchments had confirmed the existence of shale fractures as a widespread feature in this region (Guo et al., 2019; Hasenmueller et al., 2017; Lin & Zhou, 2008). The overall higher preferential flow frequency at Shale Hills (Figure 5) might also accelerate the percolation rate in the shale soils, which significantly shortened the sensors' response times (Figure 4). The higher preferential flow tendency in clay-rich soils had also been confirmed by other studies but using different methods (Djodjic, Bergstrom, Ulen, & Shirmohammadi, 1999; Glaesner, Kjaergaard, Rubaek, & Magid, 2011).

AWC had proved to be a significant control on preferential flow, and the absolute water content was sometimes negatively correlated with preferential flow occurrence (Grant et al., 2019; Hardie et al., 2011). At both Shale Hills and Garner Run, drier soils were more favourable for preferential flows to occur except for Shale Hills toe slope site. This could possibly be explained by the water repellence factor (Doerr, Shakesby, & Walsh, 2000), or the cracks usually formed in clay-rich soils under dry conditions (Simard, Beauchemin, & Haygarth, 2000). At the toe slope position, Shale Hills had a higher clay content (27.4%) than Garner Run (19.8%). However, Shale Hills tended to have more preferential flows under wet condition at this site. This indicated that hillslope position was a stronger control at the Shale Hills toe slope site than lithology, where shallow groundwater facilitated rapid wet-up of deep soils and thus ephemeral saturation at the 40-cm soil moisture sensor happened more frequently. However, the groundwater table depth was still below the deepest sensor (40 cm) based on the nearest groundwater level measured at Shale Hills (Figure S2). Thus, soil moisture change detected in our study were attributed to preferential flow events. Given the importance of AWC, seasonality could be another potential control on preferential flows, that dry periods (July–September) might result in more preferential flow occurrences. Previous studies had confirmed that soil hydraulic properties could change temporally through time (Hu, Shao, Wang, Fan, & Horton, 2009). Climate change also had combined

biological, chemical, and physical feedbacks on soil hydraulic functions (Robinson et al., 2019).

For both catchments, the mineral layer (B horizon) was a recurrent hot spot for preferential flows to occur (Table 2; Figure 7). For Garner Run, macropore flow was also developed in the eluviation layer (E horizon), which was possibly due to granular materials left by material dissolution (Hencher, 2010). “Bottom-up” type preferential flows (e.g., rapid rise-up and long-saturation state at the bottom layer during events) occurred at Shale Hills toe slope site and Garner Run ridge-top site, as was also discussed in the above paragraph. Garner Run and Shale Hills toe slope sites also represented convergent zones where more subsurface flows occurred, which was another potential hillslope position effect. At Shale Hills, the slope gradient was higher than that of Garner Run (Table 1). This would likely lead to faster lateral water movement, especially in the mid-slope and toe slope positions.

4.3 | Limitations and future studies

Through this study, we successfully measured the frequency of preferential flows in soils of steep, boulder-strewn colluvium in forested hillslopes of the mid-Appalachians using the lag analysis method. Challenges of studying process-level water movement in boulder fields still existed, as the spatial pattern of above- and below-ground boulders remained unknown. The heterogeneity within boulder fields could potentially impact the upscaling of plot-scale preferential flows to watershed-scale preferential flows. Future studies could employ a combination of plot-scale soil moisture measurements, meso-scale soil moisture measurements, and catchment-scale soil moisture measurements (Babaeian et al., 2019). Additionally, as traditional infiltrometers failed to perform in these boulder-rich, mountainous regions, more attention should be paid to developing new portable infiltrometers that capture deeper infiltration rates that would allow for more accurate flux estimation.

5 | CONCLUSION

We compared soil moisture content, percolation rate and preferential flows occurrences between a shale site (Shale Hills) and a sandstone site (Garner Run). Shale Hills had higher preferential flow frequency compared to Garner Run for all hillslope positions, while different hillslope positions revealed different levels of contrasts. The higher propensity to preferential flows at Shale Hills was likely attributed to its special soil structure, including the orientation of the shale fractures in the topsoil based on field observation. After calculating the lag time between precipitation and soil moisture, we did not find longer response times at Shale Hills, which indicated the potential roles of preferential flow in accelerating percolation rates. Over unfrozen periods, preferential flow occurrence was more likely associated with drier soil conditions, whereas significant relationship was found at three out of eight sites. We also found “bottom-up” type preferential flows for Shale Hills toe slope site and Garner Run ridge-top site. At

Shale Hills it was mainly caused by high groundwater table that formed high capillary fringes, while for Garner Run, it was possibly caused by the dense B_w layer at the 40-cm depth. The B_w horizon at Shale Hills and the E/AE horizon at Garner Run served as characteristic preferential flow pathways for each catchment.

This study highlighted the usefulness of wireless soil moisture sensor network to capture dynamic preferential flow pathways, but it also revealed the challenges of studying preferential flows in boulder-strewn fields. Lithology and soil structure should be well described when simulating percolation processes, instead of predicting soil hydrologic conductivity based only on soil texture. In addition to that, understanding the impact of lithology and climate variations on soil moisture dynamics and preferential flows could provide useful information to understand streamflow generation and solute transport for headwater catchments.

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DATA AVAILABILITY STATEMENT

The Ground HOG soil moisture data have been digitized and are accessible through national CZO data portal (<https://criticalzone.org/shale-hills/data/datasets/>) from the corresponding author (jxd523@psu.edu) upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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